County-wise weather variable rankings

Alabama PM2.5 Study | Local workbook analysis

Ten county worksheets; 365 input records per county. Each county was analyzed independently using the existing Python PCA workflow.

Main findings

Dew point ranks first in eight counties. Precipitation ranks first in Jefferson, and wind speed ranks first in Sumter. Sumter precipitation is constant and is excluded from scaled PCA.

Temperature and dew point exceed the absolute-correlation redundancy threshold of 0.90 in every county. No analyzed variable has retained communality below 0.50. These results do not justify declaring a weather variable irrelevant to PM2.5.

County	First-ranked variable	Rows used	PCs	Variance
Baldwin	Dew Point	363	3	84.27%
Clay	Dew Point	365	3	83.44%
DeKalb	Dew Point	365	3	82.75%
Etowah	Dew Point	357	4	91.45%
Jefferson	Precipitation	365	4	90.97%
Mobile	Dew Point	359	3	85.01%
Montgomery	Dew Point	357	4	91.93%
Morgan	Dew Point	362	3	80.47%
Sumter	Wind Speed	351	3	91.11%
Tuscaloosa	Dew Point	365	3	82.94%

How to read the rankings

Importance is the percentage share of retained PCA variance contributed by a variable. Variables are standardized, and enough PCs are retained to explain at least 80% of weather variance. Scores are ordered within each county, not compared as absolute effects across counties.

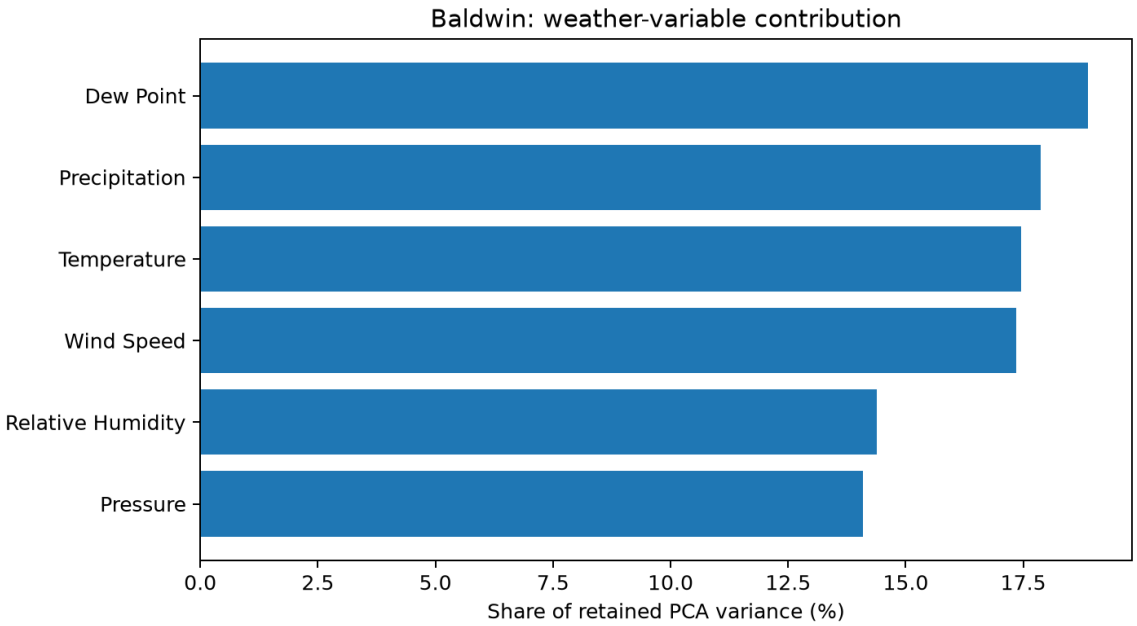
PCA contribution is not a p-value, statistical significance, or predictive importance for PM2.5. Small rank differences can be unstable. PM2.5 values are not used in this ranking.

Baldwin

Worksheet: Baldwin | 363 of 365 rows used | 2 rows omitted

3 retained PCs explain 84.27% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	18.8681	0.9540	Redundancy
2	Precipitation	17.8619	0.9032	None
3	Temperature	17.4480	0.8822	Redundancy
4	Wind Speed	17.3525	0.8774	None
5	Relative Humidity	14.3785	0.7270	None
6	Pressure	14.0910	0.7125	None



Temperature / dew point correlation: 0.9408. Both are flagged for redundancy; neither was removed for this analysis.

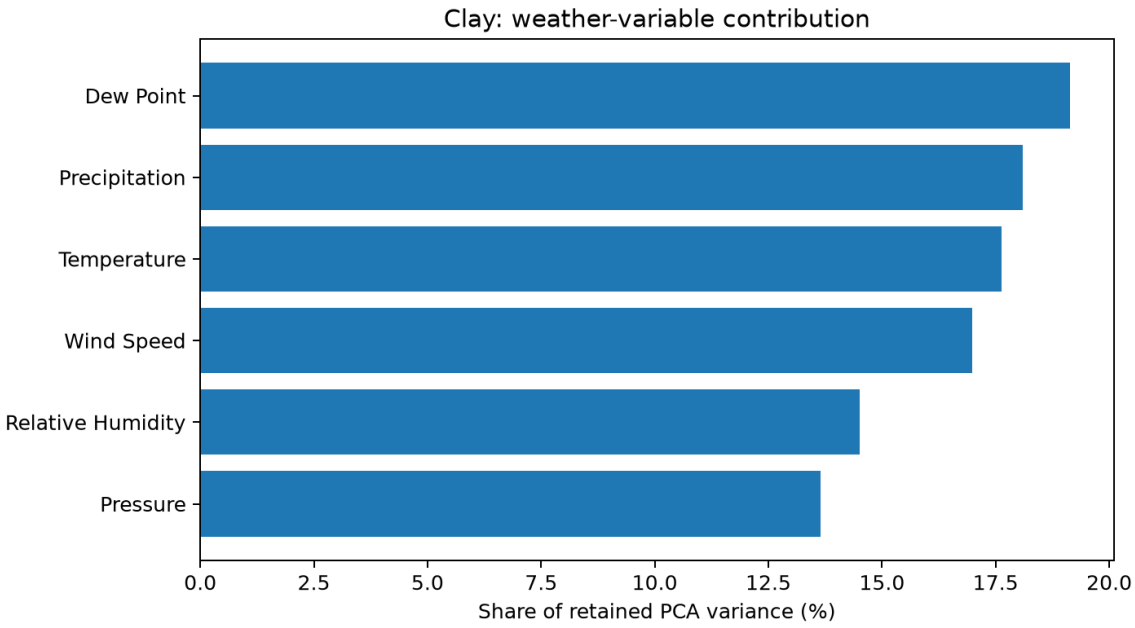
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Clay

Worksheet: Clay | 365 of 365 rows used | 0 rows omitted

3 retained PCs explain 83.44% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	19.1408	0.9583	Redundancy
2	Precipitation	18.1003	0.9062	None
3	Temperature	17.6323	0.8828	Redundancy
4	Wind Speed	16.9786	0.8501	None
5	Relative Humidity	14.5066	0.7263	None
6	Pressure	13.6414	0.6830	None



Temperature / dew point correlation: 0.9647. Both are flagged for redundancy; neither was removed for this analysis.

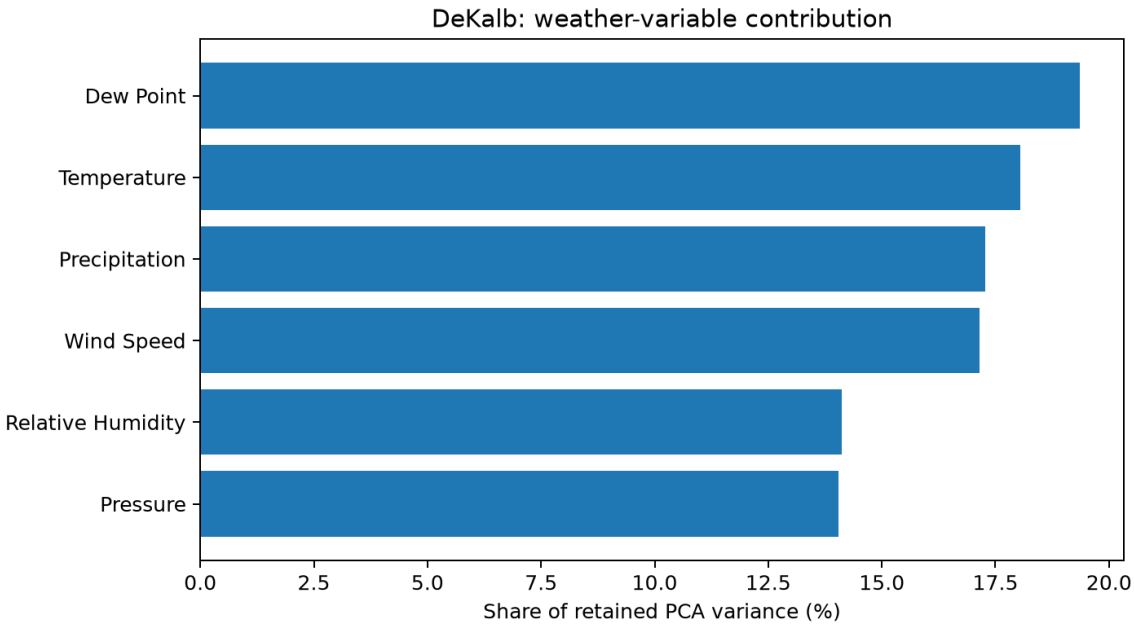
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

DeKalb

Worksheet: Dekalb | 365 of 365 rows used | 0 rows omitted

3 retained PCs explain 82.75% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	19.3593	0.9611	Redundancy
2	Temperature	18.0417	0.8957	Redundancy
3	Precipitation	17.2799	0.8579	None
4	Wind Speed	17.1591	0.8519	None
5	Relative Humidity	14.1135	0.7007	None
6	Pressure	14.0465	0.6974	None



Temperature / dew point correlation: 0.9649. Both are flagged for redundancy; neither was removed for this analysis.

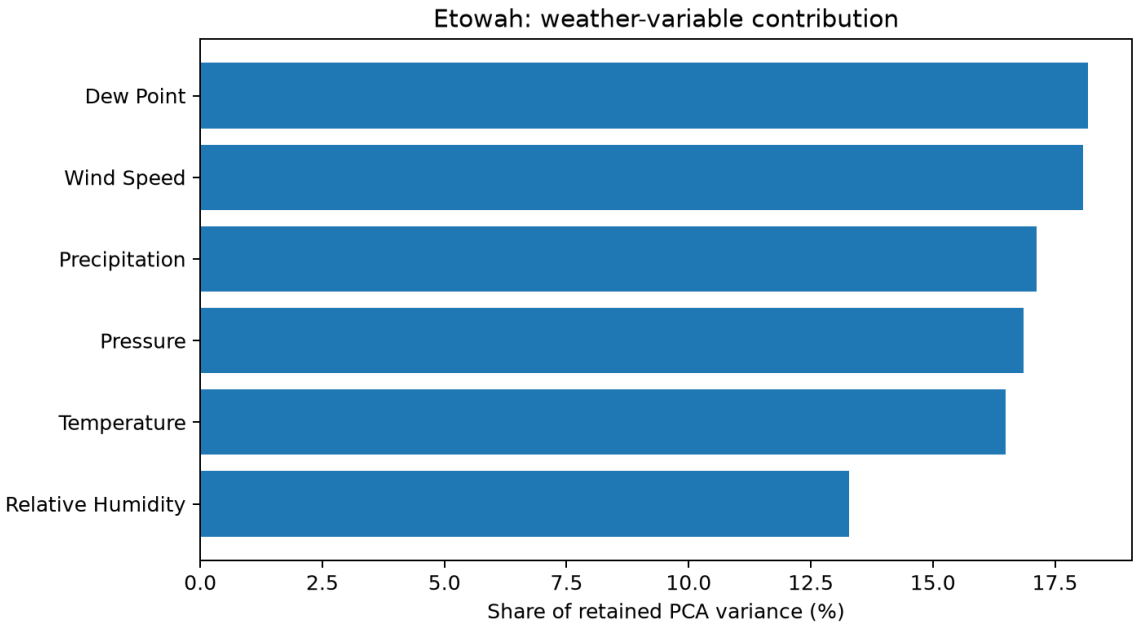
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Etowah

Worksheet: Etohaw | 357 of 365 rows used | 8 rows omitted

4 retained PCs explain 91.45% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	18.1687	0.9969	Redundancy
2	Wind Speed	18.0758	0.9918	None
3	Precipitation	17.1208	0.9394	None
4	Pressure	16.8558	0.9249	None
5	Temperature	16.4961	0.9051	Redundancy
6	Relative Humidity	13.2828	0.7288	None



Temperature / dew point correlation: 0.9281. Both are flagged for redundancy; neither was removed for this analysis.

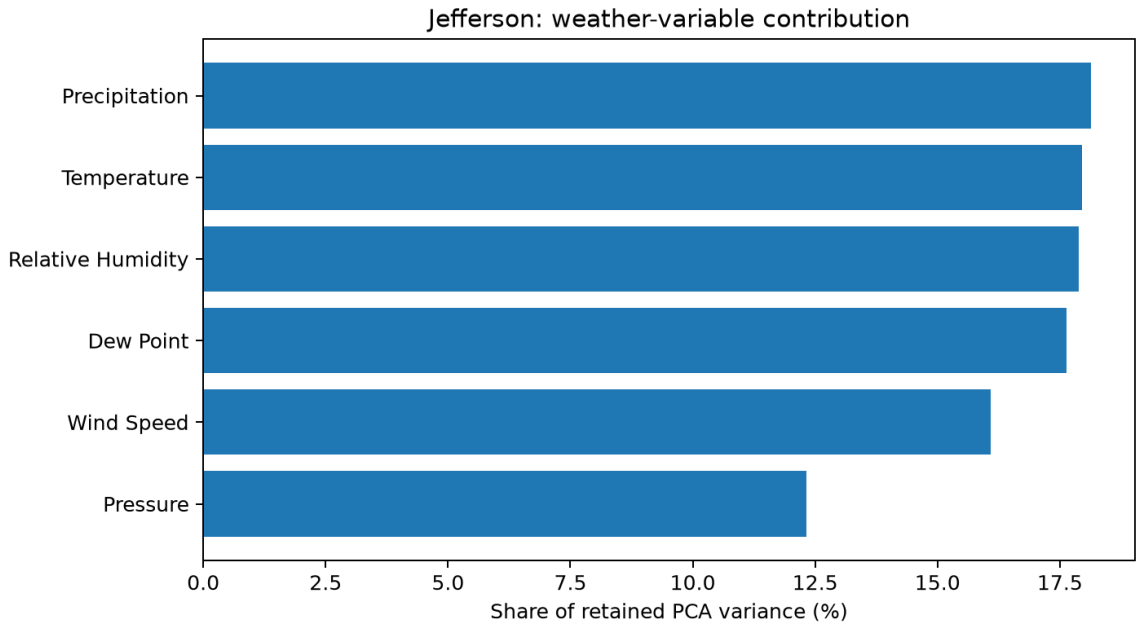
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Jefferson

Worksheet: Jefferson_NB | 365 of 365 rows used | 0 rows omitted

4 retained PCs explain 90.97% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Precipitation	18.1271	0.9895	None
2	Temperature	17.9461	0.9796	Redundancy
3	Relative Humidity	17.8891	0.9765	None
4	Dew Point	17.6339	0.9625	Redundancy
5	Wind Speed	16.0802	0.8777	None
6	Pressure	12.3235	0.6727	None



Temperature / dew point correlation: 0.9398. Both are flagged for redundancy; neither was removed for this analysis.

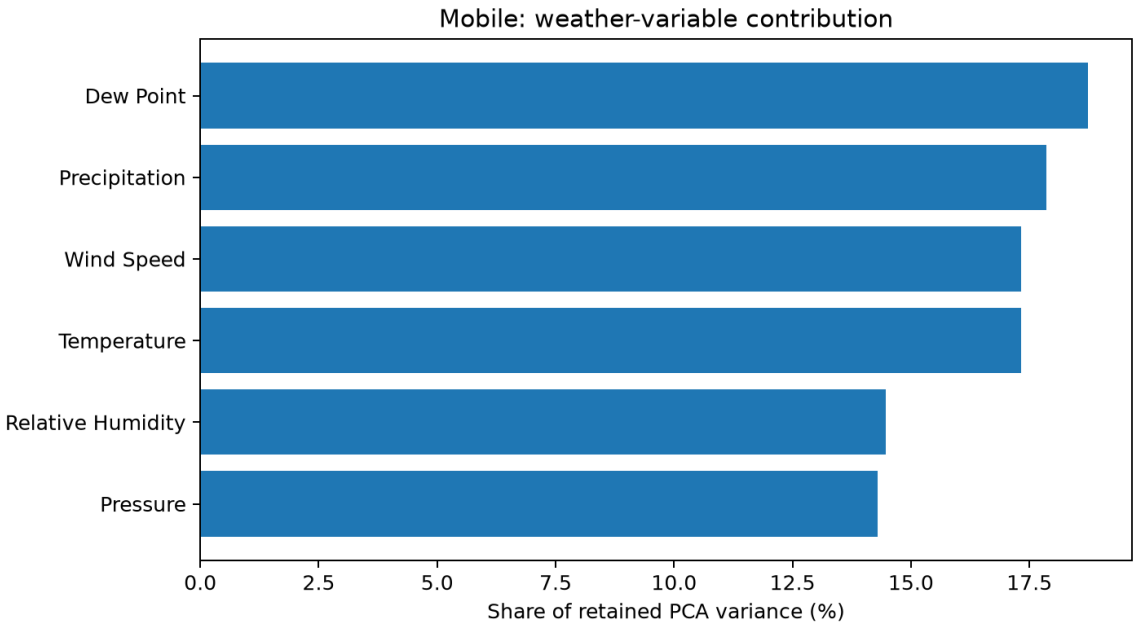
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Mobile

Worksheet: Mobile | 359 of 365 rows used | 6 rows omitted

3 retained PCs explain 85.01% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	18.7288	0.9553	Redundancy
2	Precipitation	17.8670	0.9114	None
3	Wind Speed	17.3174	0.8833	None
4	Temperature	17.3165	0.8833	Redundancy
5	Relative Humidity	14.4671	0.7379	None
6	Pressure	14.3033	0.7296	None



Temperature / dew point correlation: 0.9432. Both are flagged for redundancy; neither was removed for this analysis.

Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

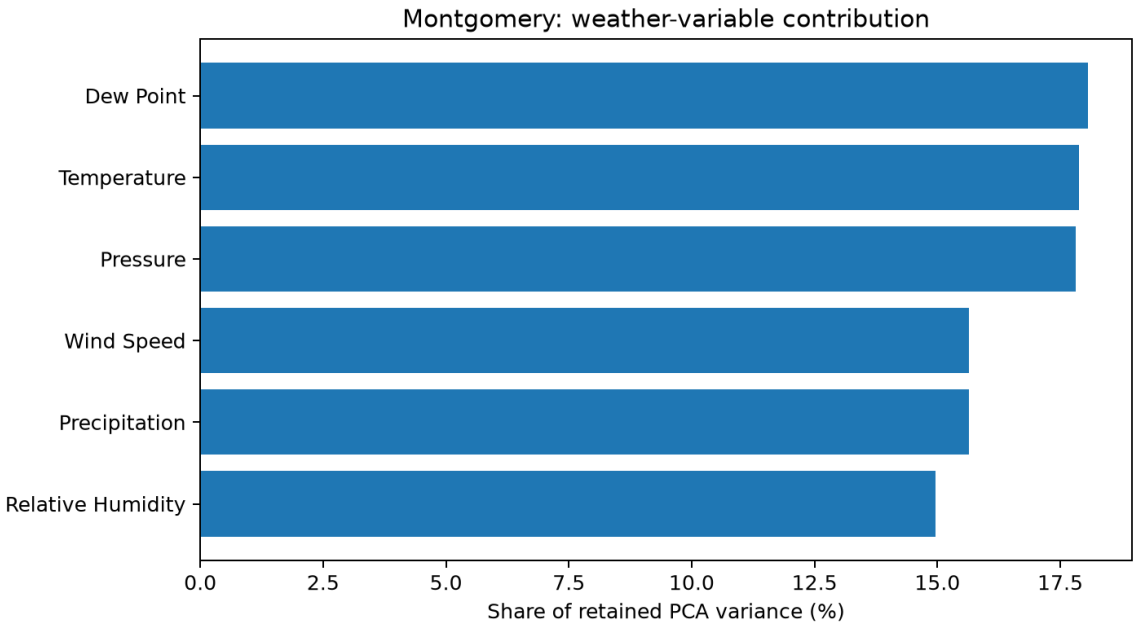
Some neighboring scores are extremely close. The numerical order is preserved, but this is not evidence of a meaningful difference in importance.

Montgomery

Worksheet: Montgomery | 357 of 365 rows used | 8 rows omitted

4 retained PCs explain 91.93% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	18.0568	0.9960	Redundancy
2	Temperature	17.8854	0.9866	Redundancy
3	Pressure	17.8200	0.9830	None
4	Wind Speed	15.6377	0.8626	None
5	Precipitation	15.6361	0.8625	None
6	Relative Humidity	14.9640	0.8254	None



Temperature / dew point correlation: 0.9434. Both are flagged for redundancy; neither was removed for this analysis.

Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

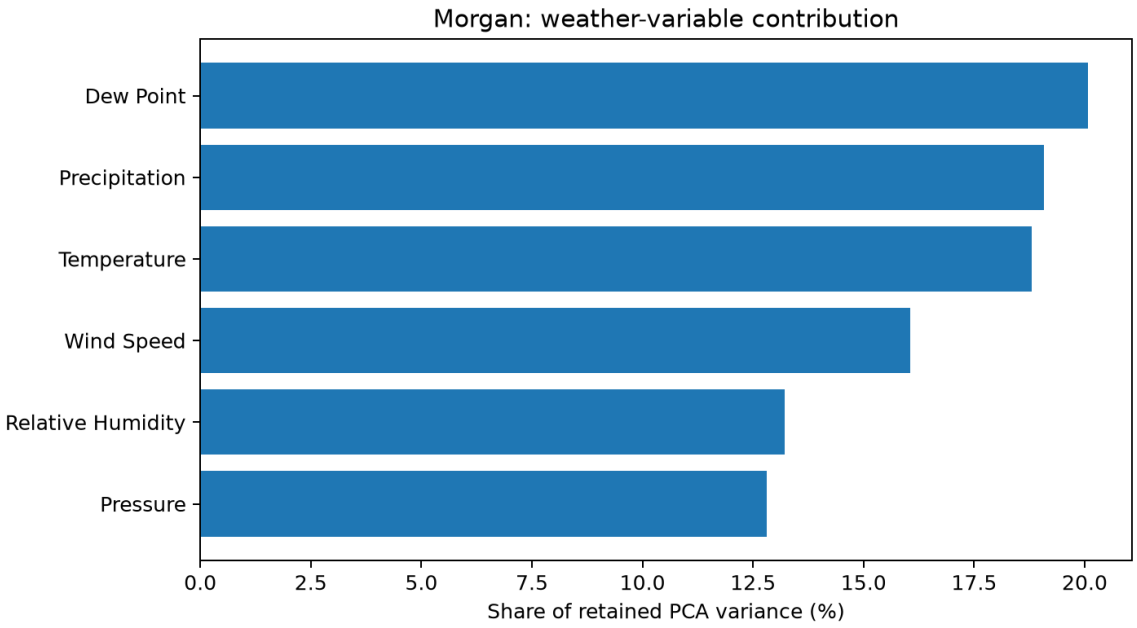
Some neighboring scores are extremely close. The numerical order is preserved, but this is not evidence of a meaningful difference in importance.

Morgan

Worksheet: Morgan | 362 of 365 rows used | 3 rows omitted

3 retained PCs explain 80.47% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	20.0660	0.9689	Redundancy
2	Precipitation	19.0710	0.9208	None
3	Temperature	18.8036	0.9079	Redundancy
4	Wind Speed	16.0458	0.7748	None
5	Relative Humidity	13.2104	0.6379	None
6	Pressure	12.8032	0.6182	None



Temperature / dew point correlation: 0.9571. Both are flagged for redundancy; neither was removed for this analysis.

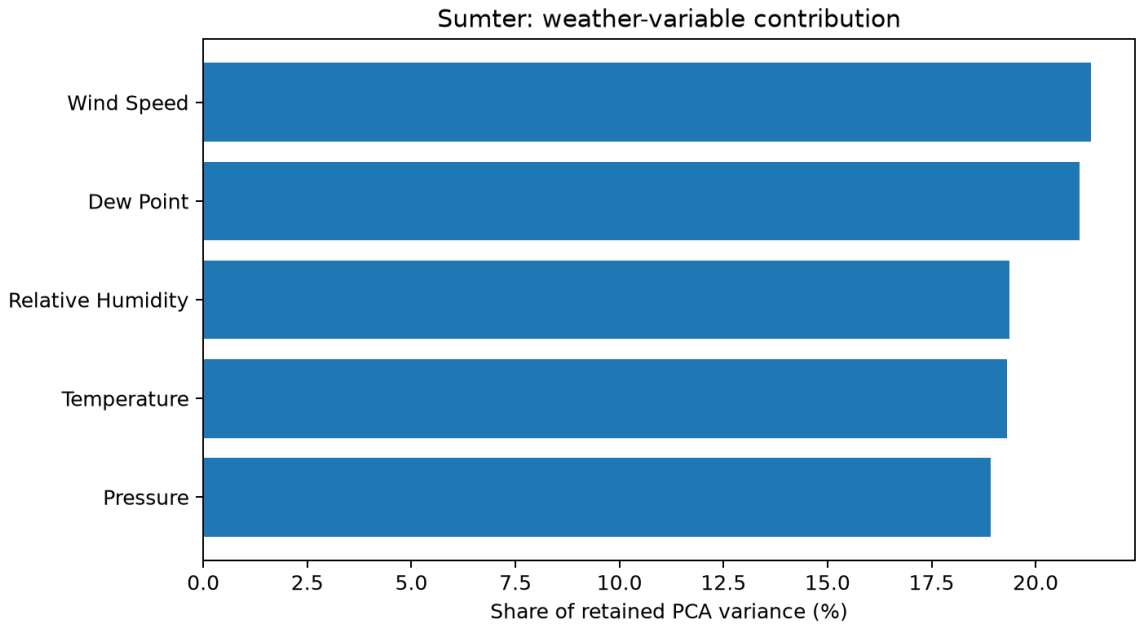
Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Sumter

Worksheet: Sumter | 351 of 365 rows used | 14 rows omitted

3 retained PCs explain 91.11% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Wind Speed	21.3211	0.9713	None
2	Dew Point	21.0603	0.9594	Redundancy
3	Relative Humidity	19.3790	0.8828	None
4	Temperature	19.3121	0.8797	Redundancy
5	Pressure	18.9274	0.8622	None
-	Precipitation	-	-	Constant / excluded



Temperature / dew point correlation: 0.9487. Both are flagged for redundancy; neither was removed for this analysis.

Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

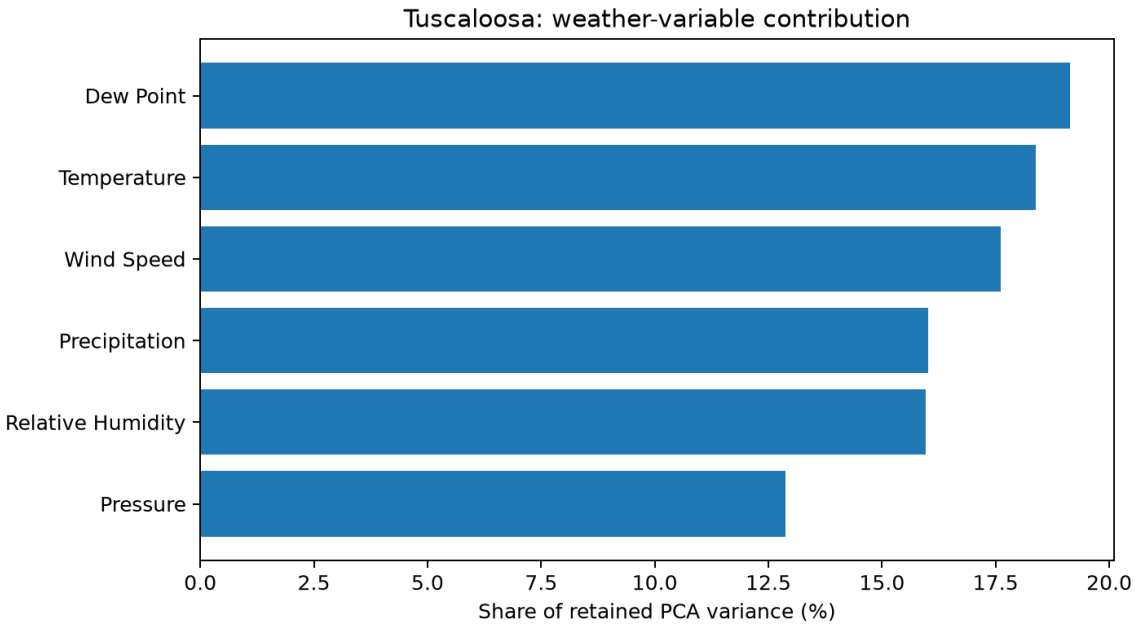
Precipitation has no variation in this workbook and is unranked. It is not assigned a zero importance score.

Tuscaloosa

Worksheet: Tuscaloosa | 365 of 365 rows used | 0 rows omitted

3 retained PCs explain 82.94% of standardized weather variance.

Rank	Variable	Importance (%)	Communality	Screening flag
1	Dew Point	19.1460	0.9527	Redundancy
2	Temperature	18.3907	0.9152	Redundancy
3	Wind Speed	17.6119	0.8764	None
4	Precipitation	16.0144	0.7969	None
5	Relative Humidity	15.9543	0.7939	None
6	Pressure	12.8826	0.6411	None



Temperature / dew point correlation: 0.9453. Both are flagged for redundancy; neither was removed for this analysis.

Communality is the fraction of a variable's standardized variance represented by the retained PCs. The low-representation flag applies below 0.50. An absent flag does not establish relevance to PM2.5.

Method and reproducibility

Input

Source workbook: PCA/Meterological data ALAbama .xlsx. All ten worksheets were matched to the county names used in the original R code. Dekalb, Etohaw and Jefferson_NB are displayed as DeKalb, Etowah and Jefferson.

Procedure

All six meteorological variables were considered in each county. All-missing and constant columns were excluded before complete-case deletion. Remaining observations were centered and scaled with sample standard deviations. Singular value decomposition produced the PCA eigenvalues and eigenvectors.

The smallest number of PCs reaching 80% cumulative variance was retained. For each variable j, communality equals the sum of eigenvalue(k) times eigenvector(j,k) squared across retained components. Importance equals 100 times that communality divided by the sum of all included variable communalities.

Flags and interpretation

Low representation: communality below 0.50. Redundancy: absolute pairwise correlation at least 0.90 on the PCA complete-case rows. Thresholds are exploratory choices. Rankings depend on the retained components and do not provide uncertainty intervals. Variables are not automatically deleted based on rank.

The existing R analysis used fixed county-specific removals. This run intentionally starts with all six weather variables, so its PCA results need not match the R benchmarks. No PM2.5 association or prediction model is fitted.

Reproduction

From the repository root, run:

```
python PCA_SR/pca_variable_ranking.py
"PCA/Meterological data ALAbama .xlsx"
--output PCA_SR/published_results
```

Join these three lines into one shell command.

Run metadata, full-precision CSV tables, loadings, correlation matrices and charts accompany this PDF in PCA_SR/published_results. The PDF builder is PCA_SR/build_report.py and requires reportlab.

Analysis environment: Python 3.14.2; NumPy 2.5.3; pandas 2.3.3.